

User manual

Restaurant review Sentiment analysis System

1. Introduction

The user manual will show a clear and starter friendly set of instructions to install, launch, and use our system. It will analyze reviews of the restaurant, and classifies them according to either positive, negative or maybe neutral depending on the sentiment analysis. After that it will display said insights via a user-friendly dashboard. This system is mostly designed keeping in mind restaurant managers, owners and even diners who want to understand their customer feedback in the easiest way possible.

2. Deployment and Installation

2.1 System Requirements

- Operating System: MacOS Monterey or later, Ubuntu 20.02+, Windows 10 or higher
- Hardware: Ideal 16gb ram (minimum 8gm ram)
- Optional yet preferred: GPU for quicker model processing
- Software: Any web browser, python 3.9+

2.2 Installation Steps

- Install latest python from python.org
- Download the project folder
- Open terminal and go to project
- Install libraries (using pip install -r requirements.txt)
- Run setup (python [setup.py](#))
- Then, start system (python [app.py](#))
- Open your local browser and go to: <http://localhost:8000>

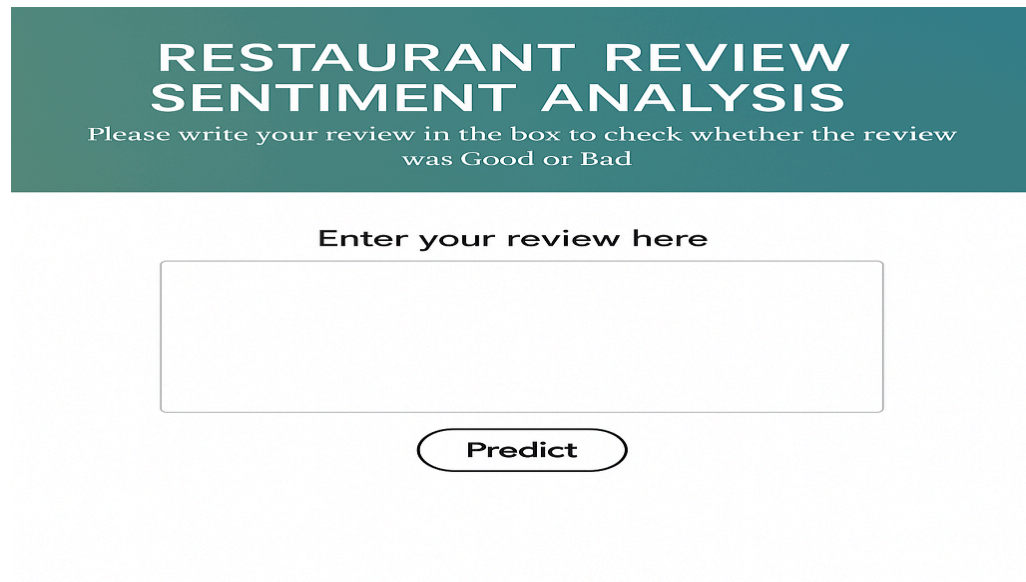
3. Main features

- This system's function is to make sentiment analysis insights both easy and accessible
- It has:
 - Automated sentiment classification to see whether reviews are being classified in the category of positive, negative or neutral
 - Insights based on aspect to highlight what it is that users are mentioning about in their reviews (that is, service, or quality of food, etc)

- Trend monitoring to show how the sentiment of the customer changes over a period of time ranging from days to even months
- Visual dashboards like charts, graphs, or word clouds that make easy to understand insights
- Customer mode to show a simplified summary for the aforementioned diners.
- Manager mode to offer a proper analysis and also filtering tools necessary for operational decisions
- Continuous learning to show that the ai model can be trained and retrained as more and more data is added, to ensure its accuracy over a long period of time

4. How to use the system walkthrough (Use this guide to be able to properly use the system)

a. Launch the Application



The screenshot shows the web interface of the 'RESTAURANT REVIEW SENTIMENT ANALYSIS' application. At the top, there is a teal header with the title in white capital letters. Below the title, a subtitle in smaller white text reads: 'Please write your review in the box to check whether the review was Good or Bad'. The main content area has a light gray background. It features a text input box with the placeholder text 'Enter your review here' centered above it. Below the input box is a rounded rectangular button with the text 'Predict' in bold.

Open the application in your browser or run the Python/Flask script (depending on how you've deployed it).

You will see the home interface with:

- A header titled "Restaurant Review Sentiment Analysis"
- A textbox area
- A Predict button

b. Enter Your Review

In the textbox labeled "Enter your review here", type any restaurant review you want the system to analyze.

Make sure your review is in normal English sentences so the AI model can interpret it correctly.

Examples:

- *“The food was amazing and the staff were very friendly!”*
- *“Terrible experience. The food was cold and service was slow.”*
- *“It was okay, nothing special.”*

c. Click the “Predict” Button

After entering the review:

- Click the Predict button
- The system will process your text and pass it into the sentiment analysis model
- The model will analyze the tone (positive or negative)

d. View the Sentiment Result

RESTAURANT REVIEW SENTIMENT ANALYSIS
Let's see whether people reviewed your restaurant Good or a Bad

Prediction: **Good Job!!! Looks like people seem to enjoy food**



Good job, seems like people
enjoy the food

Once the AI finishes prediction, you will see:

- Sentiment Output
 - Example: “**Good Job**”

- Example: “**Can do better**”
- Optionally, the system may also display:
 - Confidence score (e.g., 0.87)
 - Classification explanation
 - Highlighted keywords (if enabled)

e. Try More Reviews



The system allows unlimited predictions.

Simply:

- ★ Clear the textbox
- ★ Type a new review
- ★ Click Predict again

This helps you test multiple examples or evaluate your dataset.

f. If You Get an Error**RESTAURANT REVIEW SENTIMENT ANALYSIS**
Let's see whether people reviewed your restaurant Good or a Bad

404 Error
404
Not Found

Possible issues:

- Empty text box
- Very short input
- Non-English characters (depending on model's training)

Solution:

- Re-enter a proper review
- Refresh the page
- Make sure your backend model is running (if using local Python/Flask)

g. End of Session

When you are finished:

- Close the tab
- Or stop the local server (Ctrl + C in terminal) if you ran it directly

5. Additional walkthrough

5.1. Multiple reviews at once

- Open the application similar to that of mentioned above
- Enter multiple reviews (precisely 3)
- On click of the predict button it will provide with the outcome
- The output will be provided for the three combined reviews

RESTAURANT REVIEW SENTIMENT ANALYSIS

Please write your review in the box
To check whether the review was Good or Bad

Enter your review here

Enter your review here

Enter your review here

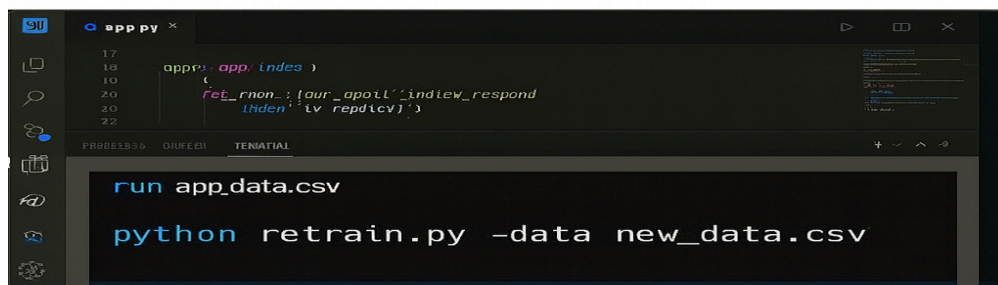
Predict

5.2. Additional walkthrough 2 - Retraining with new data

- Create a csv that has 2 columns, called review and label
- Save it
- Run `python retrain.py --data new_data.csv`
- Restart the app

new_data.csv	
review	label
Loved the drinks and ambiance	positive
The food was cold	negative

```
python retrain.py -data new_data.csv
```



The screenshot shows a JupyterLab environment. The top pane displays a Python script named `app.py` with the following code:

```
17 app = app_index()
18
19 {
20     return: {our_app: 'index_respond
21             'index' in request}
22 }
```

The bottom pane is a terminal window with the following commands:

```
run app_data.csv
python retrain.py -data new_data.csv
```